High school · olympiad thinking

The pigeonhole principle

Lesson and method

Before you begin

Euclidean division and elementary counting.

Ideas and method

If $n+1$ objects are placed in n boxes, at least one box contains two objects. Otherwise, each box would contain at most one, giving at most n objects.

More generally, if N objects occupy k boxes, one box contains at least $\lceil \frac{N}{k} \rceil$. The ceiling $\lceil x \rceil$ is the smallest integer greater than or equal to x.

Choosing the boxes is the key: remainders, birth months, small grid squares or classes of integers. Explain why every object belongs to one box.

Worked example

Among 13 people, two were born in the same month: the 12 months are the boxes. This does not guarantee two birthdays on the same day.

Explore

Choose six distinct integers from 1 to 10. Group them into pairs summing to 11 and find a complete pair.